ChemCritic 文献证据卡

PMC12007006 | Miyaura 硼化 / C – H 硼化 | 采集状态 imported_unchecked

标题	Trimethylaluminum Activates Zeolite-Confined Lanthanum Borohydrides to Enhance Catalytic C – H Borylation
作者	Li Y., Kobayashi T., Meira D., An L., Qi L., Sadow A.
出处	Journal of the American Chemical Society 2025
DOI	10.1021/jacs.4c18569
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC12007006/
许可	https://creativecommons.org/licenses/by-nc-nd/4.0/Permits non-commercial access and re-use, provided that author attribu

摘要（原文节选）

AlMe3-treatment of the borylation precatalyst La(BH4)2(THF)2.5-Ph3Si-HY30 affords La(BH4)2(AlMe3)-Ph3Si-HY30, which is the superior lanthanum-based precatalyst for benzene borylation with pinacolborane (HBpin), giving higher turnovers (>285) and improved yields (up to 42%) of phenylpinacolborane (PhBpin). Solid-state NMR spectroscopy, X-ray adsorption spectroscopy, and theoretical studies characterized the precatalytic sites in La(BH4)2(AlMe3)-Ph3Si-HY30 as κ2-O,O-{≡SiO(=Al)≡SiO}La(BH4)2(AlMe3), revealing that A ⋯

数据表 · Table 1（12 行，含数值行 12 行）

Catalytic Benzene Borylation by Organolanthanum Speciesa

catalyst	La loading (mmol/g)b	t (h)	conversion (%)c	yield (%)	turnovers
La(BH4)3(THF)3 (1)d	n.a.e	12	0	0	n.a.e
La(BH4)2(THF)2.5-Ph3Si-HY30 (2)	0.07 ± 0.02	12	97 ± 2	7.4 ± 0.4	50 ± 6
La(BH4)2(THF)x-HY30 (3)	0.10 ± 0.01	12	99 ± 1	1.8 ± 0.2	10 ± 1
La(BH4)2(AlMe3)-Ph3Si-HY30 (4)	0.07 ± 0.01	60	95 ± 3	21 ± 3.5	107 ± 6
La(BH4)2(AlMe3)-Ph3Si-HY30 (4)f	0.07 ± 0.01	108	97 ± 3	40 ± 2	13 ± 1
La(BH4)2(AlMe3)-HY30 (5)	0.09 ± 0.02	60	98 ± 2	8.1 ± 0.3	36 ± 3
La(BH4)2(THF)2.2-SiO2 (6)	0.11 ± 0.01	12	0	0	n.a.e
La(BH4)2(AlMe3)-SiO2 (7)	0.11 ± 0.02	60	96 ± 4	6.4 ± 0.4	26 ± 2
La(AlMe4)3 (8)g	n.a.e	24	51 ± 4	2.0 ± 0.3	1.2 ± 0.4
La(AlMe4)x-Ph3Si-HY30 (9)	0.07 ± 0.01	24	89 ± 3	4.1 ± 0.4	21 ± 2
La(AlMe4)x-HY30 (10)	0.11 ± 0.01	24	77 ± 4	3.8 ± 0.3	14 ± 1
La(AlMe4)x-SiO2 (11)	0.11 ± 0.02	24	70 ± 3	3.2 ± 0.4	10 ± 2

脚注（逐字）：a0.025 g of precatalyst (0.2 – 0.3 mol % La), 0.15 mL of HBpin (1.0 mmol), 0.50 mL of benzene, 120 ° C.bDetermined by inductively coupled plasma-optical emission spectroscopy (ICP-OES).cHBpin conversion.d0.003 g of La(BH4)3(THF)3 (0.007 mmol, 0.7 mol % La).eNot applicable.f0.030 g of 4, 0.01 mL of HBpi

正文对应句（逐字）：The AlMe3-treated 4 is a superior precatalyst to its precursor 2 for the reaction of benzene and HBpin to give PhBpin and H2 (Table 1). The yield of PhBpin (21 ± 4%) and TON (107 ± 6) measured for 4 are ca. 3× and 2× higher, respectively, than obtained previously using 2.29 The higher PhBpin yields,

本卡为 NCBI PMC 开放获取 XML
的逐字摘录（标题、作者、表格行列、脚注、正文句均未改写），用于参照集候选与覆盖面登记；未逐条人工核验，标记 imported_unchecked，不写入判定记录、不参与 ChemCritic 引擎的可比性判定。对外引用一律回到 DOI 原文核对页码与表号。